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**There are four modifiers used in C++. They are:**

(1) signed (2) unsigned (3) long (4) short

These four modifiers can be used with any fundamental data type. The following Table 9.6 shows the memory allocation for each data type with and without modifiers.

**Integer type**

Table 9.6 Memory allocation for Data types

| Data type | | Space in memory | | Range of value |
|---|---|---|---|---|
| | | in terms of bytes | in terms of bits | |
| **short** | short is a short name for **short int** | 2 bytes | 16 bits | -32,768 to 32,767 |
| **unsigned short** | an integer number without minus sign. | 2 bytes | 16 bits | 0 to 65535 |
| **signed short** | An integer number with minus sign | 4 bytes | 32 bits | -32,768 to 32,767 |
| **Both short and signed short are similar** | | | | |
| **int** | An integer may or may not be signed | 2 bytes | 16 bits | -32,768 to 32,767 |
| **unsigned int** | An integer without any sign (minus symbol) | 2 bytes | 16 bits | 0 to 65535 |
| **signed int** | An integer with sign | 2 bytes | 16 bits | -32,768 to 32,767 |
| **Both short and int are similar** | | | | |
| **long** | long is short name for long int | 4 bytes | 32 bits | -2147483648 to 2147483647 |
| **unsigned long** | A double spaced integer without any sign | 4 bytes | 32 bits | 0 to 4,294,967,295 |
| **signed long** | A double spaced integer with sign | 4 bytes | 32 bits | -2147483648 to 2147483647 |

The above table clearly shows that an integer type accepts only 2 bytes of data whereas a long int accepts data that is double this size i.e., 4 bytes of data. So, we can store more digits in a long int. (long is modifier and int is a fundamental data type)